

7 Serotonin is produced by neurones in the brain.

(a) How many of the following statements about serotonin are correct?

(1)

- L-dopa can be converted into serotonin in the brain
- Serotonin is a neurotransmitter
- Serotonin is produced by post-synaptic neurones

☐ **A** 0

☒ **B** 1

☒ C 2

☒ **D** 3

(b) There are 14 different serotonin receptors found in the human nervous system.

Three different genes are used to produce these receptors.

Explain how three genes can produce 14 different receptors for serotonin.

(4)

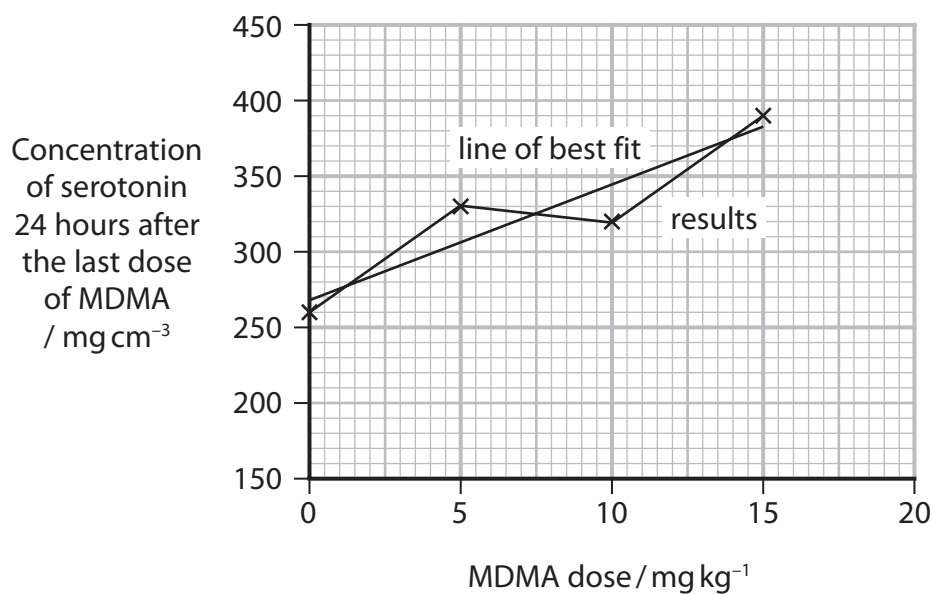
(c) The effect of MDMA (ecstasy) on serotonin concentration in rats was investigated.

Rats were given different concentrations of MDMA by mouth twice daily for a week.

The concentration of serotonin in the cerebrospinal fluid of the rats was analysed 24 hours after the last dose of MDMA.

Cerebrospinal fluid is fluid that surrounds the brain and spinal cord.

The graph shows the effect of MDMA on the concentration of serotonin and a line of best fit is shown on the graph.



- (i) Determine the effect of MDMA on the serotonin level in the cerebrospinal fluid.

(2)

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- (ii) Predict the concentration of serotonin for a dose of 17.5 mg kg^{-1} of MDMA.

Use the line of best fit.

Give your answer in g cm^{-3} .

(2)

..... g cm^{-3}

- (iii) Explain how the MDMA taken **by mouth** could affect the level of serotonin in the brain.

Use the information in the question to support your answer.

(3)

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(Total for Question 7 = 12 marks)